Customer Satisfaction (CSAT) Analytics

Project: PRDA – 02 CSAT Analytics

Tools Used: MySQL, Power BI, Excel

Dataset: CSAT Dataset

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Project Overview

The Customer Satisfaction Analytics project focuses on analysing customer service call data to understand customer satisfaction, call trends, service channels, call reasons, response time and call centre performance.

The analysis was performed using MySQL for data analysis and Power BI for visualization.

Objective

The main objective of this project is to analyse customer service data and identify patterns that can help improve customer satisfaction and service quality.

The project focuses on:

- Call volume trends
- Communication channels
- Geographic distribution
- Reasons for customer calls
- Response-time performance
- Customer sentiment
- CSAT scores
- Call centre performance

Dataset Description

The dataset contains 32,942 customer service call records and includes information such as:

- Customer ID
- Customer name
- Sentiment
- CSAT score
- Call timestamp
- Call day
- Reason for calling
- City , State
- Communication channel
- Response time
- Call duration
- Call centre

Tools Used

MySQL was used for:

- Data validation
- Record analysis
- NULL-value checking
- Duplicate checking
- Category analysis
- Aggregation and statistical analysis

Power BI was used for:

- Data transformation
- Dashboard creation
- KPI cards
- Charts
- Maps
- Interactive analysis

Data Preparation

The dataset was connected from MySQL to Power BI.

The following transformations were performed:

- Verified column data types
- Converted the call timestamp field to Date
- Checked data quality
- Checked for errors
- Verified that there were no NULL values in the analyzed fields
- Prepared the data for visualization

Key Insights

1. Overall Customer Service Volume

   The dataset contains approximately 33K customer calls. The average CSAT score is 2.07, while the average call duration is approximately 25.02 minutes.

2. Channel Distribution

   The Call-Centre channel has the highest share of calls at approximately 32.3%. Chatbot, Email and Web account for the remaining call volume.

3. Call Reasons

   Billing Questions are the most common reason for customer calls, followed by Payments and Service Outage.

4. Response Time

   The largest proportion of calls falls under the Within SLA category. Below SLA represents another substantial group, while Above SLA represents a smaller proportion.

5. Sentiment and CSAT

   Average CSAT differs considerably across sentiment categories. Very Positive calls have the highest average CSAT, while Very Negative calls have the lowest average CSAT.

6. Call Centre Performance

   Call volumes differ between call centres. Among the displayed centres, Los Angeles/CA has the highest call volume, followed by Baltimore/MD and Chicago/IL.

7. Call Duration

   The average call duration is approximately 25.02 minutes, indicating that customer-service interactions can require significant handling time.

Business Suggestions

Based on the analysis, the following suggestions can be considered:

1. Develop better self-service resources for frequently occurring billing and payment queries.
2. Investigate calls that fall into the Above SLA category to identify reasons for delayed responses.
3. Analyse Negative and Very Negative sentiment cases to identify recurring customer problems.
4. Monitor workload across high-volume call centres and adjust staffing where required.
5. Continue monitoring CSAT scores by channel, reason and response time to identify changes in customer experience.

Conclusion

The CSAT Analytics project demonstrates how customer-service data can be analysed using SQL and Power BI to identify important patterns in customer interactions.

The dashboard provides information about call volume, channels, call reasons, response time, sentiment, CSAT and call-centre activity. These insights can support data-driven improvements in customer-service operations and customer experience